

Calculus Master Guide

Integration Techniques and Differential Equations

Complete Notes with 70+ Solved Examples

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1 Integration by Parts

1.1 What is Integration by Parts?

Integration by parts is used when we have an integral of a **product** of two functions. The basic idea comes from the product rule for differentiation:

If u and v are functions of x , then:

$$\frac{d}{dx}(uv) = u \frac{dv}{dx} + v \frac{du}{dx}$$

Integrating both sides:

$$uv = \int u \frac{dv}{dx} dx + \int v \frac{du}{dx} dx$$

Rearranging gives the integration by parts formula:

$$\boxed{\int u dv = uv - \int v du}$$

1.2 The LIATE Rule (How to Choose u)

When deciding what to choose as u , use this priority list:

1. **L**ogarithmic functions: $\ln x$, $\log x$
2. **I**nverse trigonometric: $\arcsin x$, $\arctan x$
3. **A**lgebraic functions: x , x^2 , $\sqrt{x}$
4. **T**rigonometric functions: $\sin x$, $\cos x$
5. **E**xponential functions: e^x , a^x

1.3 Step-by-Step Examples

1.3.1 Example 1: Basic Case

$$\int x e^x dx$$

Solution:

1. Choose: $u = x$ (Algebraic before Exponential)
2. Then: $dv = e^x dx$
3. Differentiate u : $du = dx$
4. Integrate dv : $v = e^x$

5. Apply formula:

$$\int x e^x dx = x e^x - \int e^x dx$$

6. Simplify:

$$= x e^x - e^x + C = e^x(x - 1) + C$$

1.3.2 Example 2: Logarithmic Function

$$\int \ln x dx$$

Solution:

1. Let $u = \ln x$, $dv = dx$

2. Then $du = \frac{1}{x} dx$, $v = x$

3. Apply:

$$\int \ln x dx = x \ln x - \int x \cdot \frac{1}{x} dx = x \ln x - \int 1 dx$$

4. Final answer:

$$= x \ln x - x + C$$

1.4 20 Practice Problems with Solutions

1.4.1 Easy Level

1. $\int x dx = \frac{x^2}{2} + C$

2. $\int x e^x dx = e^x(x - 1) + C$

3. $\int x \sin x dx$

$$u = x, \quad dv = \sin x dx$$

$$du = dx, \quad v = -\cos x$$

$$\int x \sin x dx = -x \cos x - \int (-\cos x) dx$$

$$= -x \cos x + \sin x + C$$

4. $\int x \cos x dx = x \sin x + \cos x + C$

5. $\int \ln x dx = x \ln x - x + C$

1.4.2 Medium Level

6. $\int x^2 e^x dx$

First: $u = x^2, \quad dv = e^x dx$

Then: $u = 2x, \quad dv = e^x dx$

$$\int x^2 e^x dx = e^x(x^2 - 2x + 2) + C$$

7. $\int x^2 \sin x dx = -x^2 \cos x + 2x \sin x + 2 \cos x + C$

8. $\int x^3 e^x dx = e^x(x^3 - 3x^2 + 6x - 6) + C$

9. $\int x \ln x dx = \frac{x^2}{2} \ln x - \frac{x^2}{4} + C$

10. $\int \arctan x dx = x \arctan x - \frac{1}{2} \ln(1 + x^2) + C$

1.4.3 Trigonometric and Exponential Mix

11. $\int e^x \sin x dx = \frac{e^x}{2}(\sin x - \cos x) + C$

12. $\int e^x \cos x dx = \frac{e^x}{2}(\sin x + \cos x) + C$

13. $\int x e^{2x} dx = \frac{e^{2x}}{2} \left(x - \frac{1}{2}\right) + C$

14. $\int x \sin 2x dx = -\frac{x \cos 2x}{2} + \frac{\sin 2x}{4} + C$

15. $\int x \cos 3x dx = \frac{x \sin 3x}{3} + \frac{\cos 3x}{9} + C$

1.4.4 Challenging Problems

16. $\int x^2 \ln x dx = \frac{x^3}{3} \ln x - \frac{x^3}{9} + C$

17. $\int x e^{-x} dx = -e^{-x}(x + 1) + C$

18. $\int x^2 e^{-x} dx = -e^{-x}(x^2 + 2x + 2) + C$

19. $\int \ln(x^2 + 1) dx = x \ln(x^2 + 1) - 2x + 2 \arctan x + C$

20. $\int x^2 \cos 2x dx = \frac{x^2}{2} \sin 2x + \frac{x}{2} \cos 2x - \frac{1}{4} \sin 2x + C$

2 Integration by Substitution

2.1 What is Integration by Substitution?

Substitution is the reverse of the chain rule. It's used when you can spot a function and its derivative (or a constant multiple) in the integrand.

2.2 The Substitution Formula

If $u = g(x)$, then $du = g'(x)dx$, and:

$$\int f(g(x))g'(x) dx = \int f(u) du$$

2.3 Step-by-Step Examples

2.3.1 Example 1: Basic Substitution

$$\int 2x(x^2 + 1)^3 dx$$

Solution:

1. Let $u = x^2 + 1$
2. Then $du = 2x dx$
3. Substitute:

$$\int u^3 du = \frac{u^4}{4} + C$$

4. Replace u :

$$= \frac{(x^2 + 1)^4}{4} + C$$

2.3.2 Example 2: Trigonometric Substitution

$$\int \sin(3x) dx$$

Solution:

1. Let $u = 3x$, so $du = 3 dx$
2. $dx = \frac{du}{3}$
3. $\int \sin u \cdot \frac{du}{3} = \frac{1}{3}(-\cos u) + C$
4. $= -\frac{1}{3}\cos(3x) + C$

2.4 20 Practice Problems with Solutions

2.4.1 Easy Level

$$1. \int 2x(x^2 + 5) dx = \frac{(x^2+5)^2}{2} + C$$

$$2. \int \cos(3x) dx = \frac{1}{3} \sin(3x) + C$$

$$3. \int e^{4x} dx = \frac{1}{4} e^{4x} + C$$

$$4. \int \frac{2x}{x^2+1} dx = \ln(x^2 + 1) + C$$

$$5. \int \sin(5x) dx = -\frac{1}{5} \cos(5x) + C$$

2.4.2 Medium Level

$$6. \int (3x + 1)^4 dx = \frac{(3x+1)^5}{15} + C$$

$$7. \int \frac{1}{\sqrt{x+4}} dx = 2\sqrt{x+4} + C$$

$$8. \int x\sqrt{x^2 + 9} dx = \frac{1}{3}(x^2 + 9)^{3/2} + C$$

$$9. \int \tan(2x) dx = -\frac{1}{2} \ln |\cos(2x)| + C$$

$$10. \int \frac{1}{x \ln x} dx = \ln |\ln x| + C$$

2.4.3 Advanced Level

$$11. \int x\sqrt{1-x^2} dx = -\frac{1}{3}(1-x^2)^{3/2} + C$$

$$12. \int 2xe^{x^2} dx = e^{x^2} + C$$

$$13. \int \frac{x}{1+x^2} dx = \frac{1}{2} \ln(1+x^2) + C$$

$$14. \int 2x \sin(x^2) dx = -\cos(x^2) + C$$

$$15. \int \frac{1}{(5x+1)^2} dx = -\frac{1}{5(5x+1)} + C$$

2.4.4 Tricky/Exam Type

$$16. \int \frac{2x}{\sqrt{x^2+4}} dx = 2\sqrt{x^2+4} + C$$

$$17. \int \frac{3x^2}{x^3+1} dx = \ln|x^3+1| + C$$

$$18. \int \frac{\cos \sqrt{x}}{2\sqrt{x}} dx = \sin \sqrt{x} + C$$

$$19. \int \frac{1}{(x+1)\ln(x+1)} dx = \ln |\ln(x+1)| + C$$

$$20. \int xe^{x^2} dx = \frac{1}{2}e^{x^2} + C$$

3 Repeated Integration by Parts

3.1 When to Use Repeated Integration by Parts

Use repeated integration by parts when:

- You have a polynomial multiplied by an exponential or trigonometric function
- The polynomial degree is 2 or higher
- After one application, you still have a similar integral to solve

3.2 The Tabular Method (Fast Way)

For integrals of the form $\int x^n f(x) dx$ where $f(x)$ is e^{ax} , $\sin(ax)$, or $\cos(ax)$:

Differentiate	Integrate	Sign
x^n	$f(x)$	+
nx^{n-1}	$\int f(x) dx$	-
$n(n-1)x^{n-2}$	$\int \int f(x) dx^2$	+
$\vdots$	$\vdots$	$\vdots$
0	$\underbrace{\int \cdots \int}_{n+1} f(x) dx^{n+1}$	$\pm$

Multiply diagonally with alternating signs.

3.3 Examples

3.3.1 Example 1: $x^2 e^x$

$$\begin{aligned} \int x^2 e^x dx &= x^2 e^x - 2x e^x + 2e^x + C \\ &= e^x(x^2 - 2x + 2) + C \end{aligned}$$

3.3.2 Example 2: $x^3 \sin x$

$$\int x^3 \sin x dx = -x^3 \cos x + 3x^2 \sin x + 6x \cos x - 6 \sin x + C$$

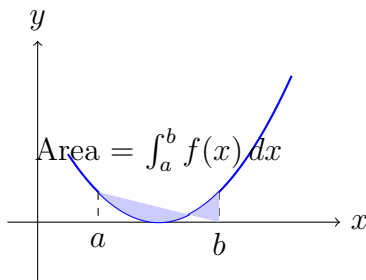
3.4 15 Practice Problems

1. $\int x e^x dx = e^x(x - 1) + C$
2. $\int x \sin x dx = -x \cos x + \sin x + C$
3. $\int x \cos x dx = x \sin x + \cos x + C$
4. $\int x^2 e^x dx = e^x(x^2 - 2x + 2) + C$
5. $\int x^2 \sin x dx = -x^2 \cos x + 2x \sin x + 2 \cos x + C$
6. $\int x^2 \cos x dx = x^2 \sin x + 2x \cos x - 2 \sin x + C$
7. $\int x^3 e^x dx = e^x(x^3 - 3x^2 + 6x - 6) + C$
8. $\int x^3 \sin x dx = -x^3 \cos x + 3x^2 \sin x + 6x \cos x - 6 \sin x + C$
9. $\int x^3 \cos x dx = x^3 \sin x + 3x^2 \cos x - 6x \sin x - 6 \cos x + C$
10. $\int x e^{2x} dx = \frac{e^{2x}}{2} \left(x - \frac{1}{2}\right) + C$
11. $\int x^2 e^{-x} dx = -e^{-x}(x^2 + 2x + 2) + C$
12. $\int x \sin 2x dx = -\frac{x \cos 2x}{2} + \frac{\sin 2x}{4} + C$
13. $\int x^2 \cos 2x dx = \frac{x^2}{2} \sin 2x + \frac{x}{2} \cos 2x - \frac{1}{4} \sin 2x + C$
14. $\int x e^{-2x} dx = -\frac{e^{-2x}}{2} \left(x + \frac{1}{2}\right) + C$
15. $\int x^2 e^{3x} dx = \frac{e^{3x}}{27}(9x^2 - 6x + 2) + C$

4 Area and Integration

4.1 The Fundamental Connection

The definite integral $\int_a^b f(x) dx$ gives the **net signed area** between $f(x)$ and the x-axis from $x = a$ to $x = b$.



4.2 Key Formulas

- **Area between curve and x-axis:** $\text{Area} = \int_a^b |f(x)| dx$
- **Area between two curves:** $\text{Area} = \int_a^b [f_{\text{top}}(x) - f_{\text{bottom}}(x)] dx$
- **Area with respect to y:** $\text{Area} = \int_c^d [x_{\text{right}}(y) - x_{\text{left}}(y)] dy$

4.3 Examples

4.3.1 Example 1: Area under $y = x^2$ from 0 to 2

$$\text{Area} = \int_0^2 x^2 dx = \left[\frac{x^3}{3} \right]_0^2 = \frac{8}{3}$$

4.3.2 Example 2: Area between $y = x$ and $y = x^2$

1. Find intersection: $x = x^2 \Rightarrow x = 0, 1$
2. Top function: $y = x$, Bottom: $y = x^2$
3. $\text{Area} = \int_0^1 (x - x^2) dx = \left[\frac{x^2}{2} - \frac{x^3}{3} \right]_0^1 = \frac{1}{6}$

4.4 15 Practice Problems

1. Area under $y = 3x$ from 0 to 4: $\int_0^4 3x \, dx = 24$
2. Area under $y = 5$ from 1 to 3: $\int_1^3 5 \, dx = 10$
3. Area under $y = x^3$ from 1 to 2: $\int_1^2 x^3 \, dx = \frac{15}{4}$
4. Area between $y = 2x$ and $y = x^2$: $\int_0^2 (2x - x^2) \, dx = \frac{4}{3}$
5. Area between $y = x^2$ and $y = 4$: $\int_{-2}^2 (4 - x^2) \, dx = \frac{32}{3}$
6. Area between $y = x^2$ and x-axis from -2 to 2: $\int_{-2}^2 x^2 \, dx = \frac{16}{3}$
7. Area enclosed by $y = x^2$ and $y = 2x$: $\frac{4}{3}$
8. Area between $y = \sin x$ and x-axis from 0 to π : $\int_0^\pi \sin x \, dx = 2$
9. Area under $y = \cos x$ from 0 to $\frac{\pi}{2}$: $\int_0^{\pi/2} \cos x \, dx = 1$
10. Total area of $y = \sin x$ from 0 to 2π : 4
11. Area between curve and y-axis: $x = y^2$, $0 \leq y \leq 1$: $\int_0^1 y^2 \, dy = \frac{1}{3}$
12. Area between $y = e^x$ and x-axis from 0 to 1: $\int_0^1 e^x \, dx = e - 1$
13. Area between $y = \ln x$ and x-axis from 1 to e : $\int_1^e \ln x \, dx = 1$
14. Area between $y = \sqrt{x}$ and $y = x$: $\int_0^1 (\sqrt{x} - x) \, dx = \frac{1}{6}$
15. Area between $y = x^3$ and $y = x$: $\int_{-1}^1 (x - x^3) \, dx = 0$ (net) but actual area = 1

5 Differential Equations

5.1 Basic Concepts

A differential equation involves derivatives of an unknown function.

- **Order:** Highest derivative present
- **Degree:** Power of the highest derivative (after rationalizing)

5.2 Types of First-Order Differential Equations

5.2.1 1. Separable Equations

Form: $\frac{dy}{dx} = f(x)g(y)$

Solution: $\int \frac{dy}{g(y)} = \int f(x) dx$

5.2.2 2. Linear Equations

Form: $\frac{dy}{dx} + P(x)y = Q(x)$

Solution using Integrating Factor: $IF = e^{\int P(x) dx}$

5.2.3 3. Exact Equations

Form: $M(x, y) dx + N(x, y) dy = 0$ where $\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$

5.3 Examples

5.3.1 Example 1: Separable

$$\frac{dy}{dx} = xy$$

Solution:

$$\int \frac{dy}{y} = \int x dx \Rightarrow \ln |y| = \frac{x^2}{2} + C \Rightarrow y = Ae^{x^2/2}$$

5.3.2 Example 2: Linear

$$\frac{dy}{dx} + y = e^x$$

Solution:

$$\begin{aligned}
 \text{IF} &= e^{\int 1 dx} = e^x \\
 e^x \frac{dy}{dx} + e^x y &= e^{2x} \\
 \frac{d}{dx}(ye^x) &= e^{2x} \\
 ye^x &= \frac{1}{2}e^{2x} + C \\
 y &= \frac{1}{2}e^x + Ce^{-x}
 \end{aligned}$$

5.4 15 Practice Problems

1. $\frac{dy}{dx} = y \Rightarrow y = Ce^x$
2. $\frac{dy}{dx} = x^2 \Rightarrow y = \frac{x^3}{3} + C$
3. $\frac{dy}{dx} = xy^2 \Rightarrow y = \frac{1}{C - \frac{x^2}{2}}$
4. $\frac{dy}{dx} = \frac{1}{y} \Rightarrow y^2 = 2x + C$
5. $\frac{dy}{dx} + y = 0 \Rightarrow y = Ce^{-x}$
6. $\frac{dy}{dx} + 2y = e^x \Rightarrow y = \frac{e^x}{3} + Ce^{-2x}$
7. $\frac{dy}{dx} - y = x \Rightarrow y = -x - 1 + Ce^x$
8. $(y + 2x) dx + x dy = 0 \Rightarrow xy + x^2 = C$
9. $(3x^2y) dx + (x^3) dy = 0 \Rightarrow x^3y = C$
10. $\frac{dy}{dx} = 4x, y(1) = 5 \Rightarrow y = 2x^2 + 3$
11. $\frac{dy}{dx} = \sin x \Rightarrow y = -\cos x + C$
12. $\frac{dy}{dx} = e^{2x} \Rightarrow y = \frac{1}{2}e^{2x} + C$
13. $\frac{dy}{dx} = \frac{x}{y} \Rightarrow y^2 = x^2 + C$
14. $\frac{dy}{dx} + xy = 0 \Rightarrow y = Ce^{-x^2/2}$
15. $\frac{dy}{dx} = y \ln y \Rightarrow \ln(\ln y) = x + C$

6 Quick Reference Guide

6.1 Integration Formulas

Type	Formula
By Parts	$\int u dv = uv - \int v du$
Substitution	$\int f(g(x))g'(x) dx = \int f(u) du$
Power Rule	$\int x^n dx = \frac{x^{n+1}}{n+1} + C, n \neq -1$
Exponential	$\int e^x dx = e^x + C$
Trigonometric	$\int \sin x dx = -\cos x + C$ $\int \cos x dx = \sin x + C$
Logarithmic	$\int \frac{1}{x} dx = \ln x + C$

6.2 Differential Equations Summary

Type	Form	Solution Method
Separable	$\frac{dy}{dx} = f(x)g(y)$	Separate and integrate
Linear	$\frac{dy}{dx} + P(x)y = Q(x)$	Integrating factor
Exact	$M dx + N dy = 0, M_y = N_x$	Find potential function

6.3 Area Formulas

- Net area: $\int_a^b f(x) dx$
- Actual area: $\int_a^b |f(x)| dx$
- Between curves: $\int_a^b [f_{\text{top}} - f_{\text{bottom}}] dx$
- With respect to y: $\int_c^d [x_{\text{right}} - x_{\text{left}}] dy$

Final Exam Tips

- For product of functions: Try integration by parts (LIATE)
- For composite functions: Try substitution
- For polynomial $\times$ exp/trig: Use repeated parts or tabular method
- Always +C for indefinite integrals Check if function crosses x-axis when finding area